L5b: Minimum-Variance Portfolios, the Efficient Frontier, and the Capital Allocation Line

CHEME 5660 Cornell University

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Objectives for Today

Today we take the estimated growth-rate means and covariance of L5a and ask how much of our wealth to put in each asset: minimum-variance portfolios, the efficient frontier, and what a risk-free asset does to the choice.

Today's concepts

- Formulate portfolio reward and risk: the expected growth rate and growth-rate variance of a portfolio as functions of $\mathbf{w}$, $\bar{\mathbf{g}}$, and Σ_g, and what the one-period objective does and does not describe.
- Construct minimum-variance portfolios and the frontier: the global minimum-variance portfolio in closed form, the long-only quadratic program, and the feasible set, the frontier, and its efficient branch.
- Add a risk-free asset: the capital allocation line, the tangent portfolio, two-fund separation and what breaks it, and when the tangent portfolio is the market portfolio.

Lectures and Examples

Lecture:
CHEME-5660-L5b-Lecture-MAGBM-Data-Portfolios-Fall-2026.ipynb

Example 1:
CHEME-5660-L5b-Example-Data-MinVar-Portfolio-Fall-2026.ipynb

Example 2:
CHEME-5660-L5b-Example-MAGBM-Portfolio-Fall-2026.ipynb

The first example estimates the inputs for a chosen set of firms, computes the minimum-variance portfolios and the efficient frontier, finds the tangent portfolio and the capital allocation line, and checks the optimized portfolios on 2025 data the optimizer never saw. The second simulates the minimum-variance allocation with correlated MAGBM paths and evaluates its scaled NPV.

Concept Review: MAGBM and the Covariance Matrix

L5a: M assets with mean growth rates $\bar{g}_i$, covariance rate $\mathbf{C}$ with factor $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$, one shock vector $\mathbf{Z}$ per step shared by all:

$$S_i(t+\Delta t) = S_i(t)\exp\Big[\bar{g}_i\,\Delta t + \sqrt{\Delta t}\,(\mathbf{A}\mathbf{Z})_i\Big], \qquad \mathbf{g} = \bar{\mathbf{g}} + \tfrac{1}{\sqrt{\Delta t}}\mathbf{A}\mathbf{Z} \implies \mathrm{Cov}(\mathbf{g}) = \tfrac{\mathbf{C}}{\Delta t}.$$

- From N days of data: sample means $\mathbf{g}'$ (estimate of $\bar{\mathbf{g}}$), $\hat{\Sigma}_g = \tilde{\mathbf{G}}^\top\tilde{\mathbf{G}}/(N-1)$ with $\tilde{\mathbf{G}}$ the centered growth-rate data matrix, $\hat{\mathbf{C}} = \Delta t\,\hat{\Sigma}_g$, $\rho_{ij} = \hat{\Sigma}_{g,ij}/\sqrt{\hat{\Sigma}_{g,ii}\hat{\Sigma}_{g,jj}}$.
- Positive semidefinite by construction: $\mathbf{v}^\top\hat{\Sigma}_g\mathbf{v} = \|\tilde{\mathbf{G}}\mathbf{v}\|_2^2/(N-1) \geq 0$, so today's minimization is convex; $N \gg M$ makes it positive definite.

Catch-up: ../L5a/CHEME-5660-L5a-Example-CovarianceMatrix-Fall-2026.ipynb (Task 2, if not reached in L5a)

Concept Review: Portfolio Wealth and the Trade Rule

For initial wealth W_0 and weights $\mathbf{w}$ on the long-only simplex ($w_i \geq 0$, $\sum_i w_i = 1$), buy $n_i = w_i W_0 / S_i(0)$ shares and hold them:

$$W_t = \sum_{i \in \mathcal{P}} n_i S_i(t), \qquad \rho_T = \frac{W_T}{W_0}\, e^{-g_b T} - 1,$$

with g_b the benchmark growth rate and T the holding period (L4b's scaled NPV, with wealth in place of the share position). A target $\rho_\star$ turns it into a probability, estimated by simulation for a portfolio because a sum of correlated lognormals has no closed form.

L5a explored $\mathbf{w}$ with Dirichlet draws and preferred no point of the simplex. Today we optimize.

Modern Portfolio Theory: Portfolio Reward

Markowitz (1952): an asset is not judged in isolation; what matters is how its growth rate moves with the rest of the portfolio. Take M risky assets held at weights $\mathbf{w}$ ($\sum_i w_i = 1$) over one period, with growth rates $\mathbf{g}$, mean $\bar{\mathbf{g}} = \mathbb{E}[\mathbf{g}]$, and covariance $\Sigma_g = \mathrm{Cov}(\mathbf{g})$. The portfolio growth rate over the period is the weighted growth rate, so its reward is:

$$g_p = \mathbf{w}^\top \mathbf{g} \quad \Longrightarrow \quad \boxed{\mathbb{E}[g_p] = \sum_{i \in \mathcal{P}} w_i \, \mathbb{E}[g_i] = \mathbf{w}^\top \bar{\mathbf{g}}}$$

estimated from data by $\mathbf{w}^\top \mathbf{g}'$; units inverse years.

Reward is linear in the weights. Risk is not, and that asymmetry is the whole subject.

What the Objective Is and Is Not

- One period at weights $\mathbf{w}$: exact simple return $R_p = \sum_i w_i (e^{g_i \Delta t} - 1)$, exact log growth rate $\Delta t^{-1} \ln(\sum_i w_i e^{g_i \Delta t})$; to first order in Δt both give $\mathbf{w}^\top \mathbf{g}$. So $g_p = \mathbf{w}^\top \mathbf{g}$ is the linearized one-period growth rate.
- Over many periods, the log growth of buy-and-hold wealth W_t is not the weighted sum of the assets' log growth rates: the weights drift, and the logarithm of a sum is not the sum of logarithms (L5a).
- MPT is a statement about this one-period statistic. The frontier, the tangent portfolio, and the capital allocation line all inherit that scope.
- Example 1 therefore evaluates the optimized weights afterwards with the buy-and-hold wealth rule: an exact, and different, calculation.

Modern Portfolio Theory: Portfolio Risk

The risk of the portfolio is the variance of its growth rate, which collects every asset variance and every pairwise covariance:

$$\boxed{\mathrm{Var}(g_p) = \sum_{i\in\mathcal{P}}\sum_{j\in\mathcal{P}} w_i w_j \underbrace{\mathrm{Cov}(g_i, g_j)}_{=\rho_{ij}\sqrt{\Sigma_{g,ii}\Sigma_{g,jj}}} = \mathbf{w}^\top \Sigma_g \mathbf{w}}$$

with $\sigma_{g,p} = \sqrt{\mathrm{Var}(g_p)}$ the portfolio's growth-rate standard deviation (inverse years; the volatility would be $\sigma_{g,p}\sqrt{\Delta t}$). Two assets at weights w and $1-w$, $0 < w < 1$:

$$\mathrm{Var}(g_p) = w^2\Sigma_{g,11} + (1-w)^2\Sigma_{g,22} + 2w(1-w)\,\rho_{12}\sqrt{\Sigma_{g,11}\Sigma_{g,22}}.$$

At $\rho_{12} = 1$ a perfect square: mixing buys nothing. For every $\rho_{12} < 1$ the standard deviation falls below the weighted average; if ρ_{12} is below the ratio of the smaller to the larger standard deviation, some mixture beats either asset alone. Diversification is a statement about covariance, not about the number of holdings.

Growth Rates versus Returns

Everything can be written with log returns $\mathbf{r} = \Delta t\, \mathbf{g}$ instead: $\mathbb{E}[r_p] = \Delta t\, \mathbb{E}[g_p]$ (dimensionless), $\mathrm{Var}(r_p) = \Delta t^2\ \mathrm{Var}(g_p)$, and $\Sigma_r = \Delta t^2 \Sigma_g$ (L5a's scaling rule).

- The optimizer does not care: multiplying the covariance by any positive constant, and the means and target by any (other) positive constant, leaves the minimizing weights unchanged.
- Every reported number does change, so the units of the risk axis, the growth axis, and the Sharpe ratio must be stated.
- Daily data: $\hat{\Sigma}_g = 252\, \hat{\mathbf{C}}$. Growth-rate variances look large next to annual covariance-rate entries; that factor is the unit conversion, not an estimation failure.

The Global Minimum-Variance Portfolio

No growth requirement at all: the fully invested portfolio with the smallest variance. Shorts allowed, budget constraint only, Σ_g symmetric positive definite, $\mathbf{1} \in \mathbb{R}^M$ the vector of ones:

$$\boxed{\min_{\mathbf{w}} \ \tfrac{1}{2}\mathbf{w}^\top \Sigma_g \mathbf{w} \ \text{ s.t. } \ \mathbf{1}^\top \mathbf{w} = 1 \quad \Longrightarrow \quad \mathbf{w}_{\mathrm{GMV}} = \frac{\Sigma_g^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma_g^{-1}\mathbf{1}}, \qquad \sigma^2_{g,\mathrm{GMV}} = \frac{1}{\mathbf{1}^\top \Sigma_g^{-1}\mathbf{1}}}$$

- The one half is a convenience; it cancels in the derivation and does not move the minimizer.
- Nothing keeps the weights non-negative. In Example 1 the closed form shorts several volatile names to hedge the others.
- Depends on Σ_g only, not on $\bar{\mathbf{g}}$: the reason it is a standard baseline (last frames).

Derivation: The Lagrangian

Introduce a multiplier $\lambda \in \mathbb{R}$ for the budget constraint:

$$\mathcal{L}(\mathbf{w}, \lambda) = \tfrac{1}{2}\mathbf{w}^\top \Sigma_g \mathbf{w} - \lambda(\mathbf{1}^\top \mathbf{w} - 1).$$

- Stationarity: $\Sigma_g \mathbf{w} - \lambda \mathbf{1} = \mathbf{0}$, so $\mathbf{w} = \lambda \Sigma_g^{-1} \mathbf{1}$.
- Budget: $\mathbf{1}^\top \mathbf{w} = 1$ gives $\lambda = (\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1})^{-1}$ and the boxed weights.
- Substituting back: $\mathbf{w}^\top \Sigma_g \mathbf{w} = \lambda^2\, \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1} = \lambda$, the boxed variance.
- $\Sigma_g \succ 0$ makes the objective strictly convex, so the stationary point is the unique minimum. ■

The Long-Only Problem

Add $w_i \geq 0$ for all i (with the budget constraint, no leverage either): a convex quadratic program, positive semidefinite quadratic objective, linear constraints, minimizer possibly on a boundary with weights exactly zero. No closed form; the course package solves it in JuMP with an interior-point method:

$$\boxed{\begin{array}{rl} \text{minimize} & \mathbf{w}^\top \hat{\Sigma}_g \mathbf{w} \\ \text{subject to} & \hat{\mathbf{g}}^\top \mathbf{w} \geq g_\star, \qquad \mathbf{1}^\top \mathbf{w} = 1, \qquad 0 \leq w_i \leq 1 \ \ \forall i \in \mathcal{P} \end{array}}$$

with $\hat{\mathbf{g}}$ the estimated mean growth rates and $g_\star$ the investor's target growth rate, entering as a floor.

Set $g_\star$ at or below the smallest single-asset mean growth rate and every long-only portfolio meets the floor: the long-only GMV portfolio, whose variance can never beat the closed form's (smaller feasible set). Position limits, turnover, lots, taxes, and costs enter the same way.

The Target-Growth Problem and the Frontier

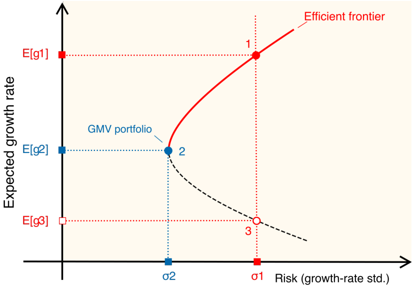
Classical form: an equality target $\bar{\mathbf{g}}^\top \mathbf{w} = g_\star$; the minimum-variance portfolios indexed by $g_\star$ form the minimum-variance frontier. Shorts allowed, with $a = \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}$, $b = \mathbf{1}^\top \Sigma_g^{-1} \bar{\mathbf{g}}$, $c = \bar{\mathbf{g}}^\top \Sigma_g^{-1} \bar{\mathbf{g}}$, $d = ac - b^2$ (positive unless all means are equal):

$$\sigma_g^2(g_\star) = \frac{a\, g_\star^2 - 2b\, g_\star + c}{d},$$

a hyperbola in the (σ_g, g) plane with vertex at the GMV portfolio ($g_\star = b/a$, $\sigma_g^2 = 1/a$).

- Efficient frontier: the upper branch, at or above the GMV growth rate; nothing feasible has at least as much growth and no more variance with one strict improvement. The lower branch is dominated by the point directly above it.
- Every fully invested $\mathbf{w}$ lands on or to the right of the hyperbola (the attainable region); long-only, the frontier lies on or to the right of it and ends at the best single asset.

The Frontier from Data: What the Floor Returns



With the target as a floor, every $g_\star$ below the GMV growth rate returns the GMV portfolio (2) and every $g_\star$ above it the efficient portfolio at that growth rate (1); the sweep never returns the lower branch (3). The weights along the sweep show which assets carry the frontier.

Example 1: CHEME-5660-L5b-Example-Data-MinVar-Portfolio-Fall-2026.ipynb

Adding a Risk-Free Asset: The Capital Allocation Line

A horizon-matched risk-free asset (a Treasury bill or zero-coupon STRIP held to maturity) has known growth rate g_f, zero variance, and zero covariance with everything. Put a fraction w_f in it and $1 - w_f$ in a risky portfolio p with $\mathbb{E}[g_p]$ and $\sigma_{g,p} > 0$; the complete portfolio c has:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)\big(\mathbb{E}[g_p] - g_f\big), \qquad \sigma_{g,c} = |1 - w_f|\, \sigma_{g,p}.$$

For $w_f \leq 1$, eliminate w_f ($1 - w_f = \sigma_{g,c}/\sigma_{g,p}$):

$$\boxed{\mathbb{E}[g_c] = g_f + \underbrace{\frac{\mathbb{E}[g_p] - g_f}{\sigma_{g,p}}}_{\text{Sharpe ratio } \mathrm{SR}_p} \sigma_{g,c}}$$

A straight line from $(0, g_f)$ through p: $0 < w_f < 1$ lends, $w_f = 0$ is p, and $w_f < 0$ borrows at g_f. A higher borrowing rate kinks the line at p.

Which Sharpe Ratio?

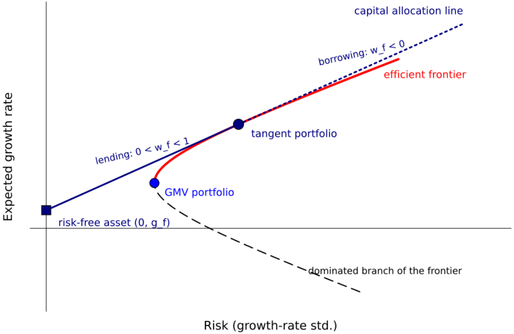
- With $\sigma_{g,p}$ in the denominator, SR_p is dimensionless (inverse years over inverse years) but is the Sharpe ratio of a one-observation-interval holding period, because $\sigma_{g,p}$ is the standard deviation of a one-day growth rate. Values are small.
- Practice divides by the volatility $\sqrt{\mathbf{w}^\top \mathbf{C}\mathbf{w}} = \sqrt{\Delta t}\,\sigma_{g,p}$ instead: the annualized Sharpe ratio $\mathrm{SR}_p / \sqrt{\Delta t}$, larger by $\sqrt{252} \approx 15.9$ for daily data.
- The two conventions rank portfolios identically, so every conclusion today holds under either. Quote which one. L6b onward quotes the annualized value; Example 1 prints both.

The Tangent Portfolio

Every risky portfolio has a CAL; steeper is better. The tangent portfolio $\mathcal{T}$ maximizes the Sharpe ratio; its CAL touches the efficient branch at $\mathcal{T}$. Shorts allowed, $\Sigma_g \succ 0$, and $\mathbf{1}^\top\Sigma_g^{-1}(\bar{\mathbf{g}} - g_f\mathbf{1}) > 0$ (GMV beats g_f; otherwise the formula gives the minimum-Sharpe portfolio):

$$\boxed{\mathbf{w}_{\mathcal{T}} = \frac{\Sigma_g^{-1}(\bar{\mathbf{g}} - g_f\mathbf{1})}{\mathbf{1}^\top\Sigma_g^{-1}(\bar{\mathbf{g}} - g_f\mathbf{1})}}$$

Long-only: no closed form; Example 1 takes the largest Sharpe ratio on the computed efficient branch.



Two-Fund Separation

- Because the tangent portfolio's CAL dominates every other, all mean–variance investors with the same inputs and the same g_f hold the same risky fund $\mathcal{T}$ and differ only in w_f: the investment decision is separated from the financing decision (Tobin, 1958).
- Exact for the unconstrained problem with a single lending and borrowing rate; a long-only constraint on the risky weights alone keeps it, with the constrained maximum-Sharpe portfolio as the fund.
- No-borrowing and position-limit constraints, or a borrowing rate above the lending rate, kink the attainable boundary; then the best risky portfolio depends on where the investor sits on the line, and separation holds piecewise, not globally.

When Is the Tangent Portfolio the Market Portfolio?

- CAPM assumptions: same beliefs about $\bar{\mathbf{g}}$ and Σ_g, same horizon, lend and borrow at one risk-free rate, every asset tradable without taxes or frictions.
- Then every investor holds the same tangent portfolio, and market clearing makes it the market portfolio: all risky assets weighted by market value. Its CAL is the capital market line; a CAL belongs to any risky portfolio, the CML to that one under those assumptions.
- An optimizer on a ticker list and eleven years of estimates returns a sample-dependent tangent portfolio for that universe and model; a broad index is one asset class and omits many risky assets.
- An equilibrium conclusion, not a label for every maximum-Sharpe solution. Example 1 compares two portfolios; it does not test the CAPM.

Estimation Risk Can Dominate Optimization

- Both closed forms carry Σ_g^{-1}, which amplifies the least-certain directions; the tangent portfolio also carries $\bar{\mathbf{g}} - g_f\mathbf{1}$, the noisiest input (two estimators of $\bar{g}_i$ from the same data differ by over ten percentage points per year for some firms), and answers with positions beyond the budget.
- Minimum-variance weights do not depend on $\bar{\mathbf{g}}$: more stable, a standard baseline next to equal weights.
- Fit on pre-decision data, test on data the fit never saw; one split is a check, not a validation. Report shrinkage, constraints, and robust objectives: they change the problem.
- MAGBM is a scenario engine for a fixed allocation, not independent validation when the same data fed optimizer and simulator.

Example 2: CHEME-5660-L5b-Example-MAGBM-Portfolio-Fall-2026.ipynb

Optional Advanced Material

Advanced 1: advanced/frontier-geometry/
CHEME-5660-L5b-Advanced-FrontierGeometry-Fall-2026.ipynb

Advanced 2: advanced/estimation-risk/
CHEME-5660-L5b-Advanced-EstimationRisk-Fall-2026.ipynb

Optional, not prerequisites for L6a. The first derives the closed-form frontier for every target, checks it against a solver, shows that two fixed portfolios generate the whole frontier (the two-fund theorem), and measures what a short-sale limit and a long-only constraint cost; the second resamples the 2014 to 2024 growth rates to see how far the frontier and the optimal weights move, traces the tangent portfolio's instability to the mean growth rates, and pushes every resampled portfolio through 2025.

Notebooks: lectures/week-5/L5b/advanced, also linked from the lecture notebook.

Summary

Today we defined a portfolio's reward and risk from its weights, means, and covariance, derived the minimum-variance portfolios and the efficient frontier, and added a risk-free asset to obtain the capital allocation line and the tangent portfolio.

Key takeaways

- Covariance determines diversification: reward is linear in the weights, variance collects every pairwise covariance; the objective is a one-period statistic.
- The frontier is a family of constrained minimizations: closed forms when shorts are allowed; long-only, a convex QP whose floor target returns the GMV portfolio and the efficient branch.
- A risk-free asset selects one risky fund: the CAL slope is the Sharpe ratio, the tangent portfolio maximizes it, and the market portfolio needs the CAPM assumptions.

Next: a single-index model in place of the full covariance matrix.

Today: two estimates, one quadratic program, and the line that a risk-free asset draws through it.